Sahaj Bhadja

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Summary

Backend Software Engineer with 2+ years building **production APIs**, **asynchronous pipelines**, secure multi-tenant services, and data-intensive systems using **Python**, **FastAPI**, SQL, AWS, Docker, testing, and CI/CD.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Software Engineering, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Data Retrieval, Operating Systems, Software Engineering

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, California*

- Architected a multi-tenant platform using Azure SQL and Blob Storage with row-level security and normalized schemas for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds.
- Built FastAPI integrations across PostgreSQL, SQL Server, and cloud storage, processing 5K+ monthly jobs with validation, reconciliation, error handling, and automated CI/CD releases sustaining 99%+ production reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, California*

- Designed concurrent processing and secure multi-format ingestion through REST APIs and AWS RDS, executing 1K+ files per batch with fault-tolerant recovery, usage tracking, and stable performance under load.
- Built Python services and responsive React interfaces for uploads, history, configuration, and live feedback, reducing user turnaround 60% while supporting reliable production workflows used by 150+ instructors.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer, SecureSync *India*

- Designed object-oriented API integrations and AWS Glue transformations for JSON, CSV, and Parquet data, improving response times 30% and pipeline efficiency 68% while reducing technical debt 20%.
- Automated nightly builds, installations, unit tests, integration tests, and reports through Jenkins, validating reliable cross-platform synchronization while collaborating directly with Microsoft and GitHub stakeholders throughout Agile delivery.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern, VibrantChem *India*

- Implemented eight MERN microservices for authentication, permissions, billing, resources, customers, leads, and notifications, reliably synchronizing 2M+ records through REST APIs and event-driven WebSocket updates across production workflows.
- Built responsive React workflows and automated Docker deployments through Jenkins CI/CD, achieving 95%+ build validation while integrating operational monitoring and analytics across controlled staging and production environments.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built cross-platform mobile and web booking workflows with Flask APIs, Firebase synchronization, payments, and webhooks, processing hourly Databricks micro-batches across 50K+ records with consistently reliable sub-200-millisecond responses.
- Designed end-to-end workflows, mapped 12+ integration points, and presented technical solutions to client stakeholders, reducing design errors 25% while supporting effective collaborative cross-functional implementation planning and alignment.

Skills

Backend & APIs: Python, FastAPI, Flask, Node.js, Express.js, Java, REST APIs, Microservices, WebSockets, Webhooks, JSON
Data & Security: PostgreSQL, SQL Server, MySQL, MongoDB, Redis, Firebase, SQL, Data Modeling, Authentication, Authorization, Row-Level Security
Cloud & Infrastructure: AWS, RDS, S3, AWS Glue, Azure, GCP, Docker, Kubernetes, Jenkins, CI/CD, Linux
Engineering: Git, GitHub, Unit Testing, Integration Testing, Debugging, Monitoring, Performance Tuning, Agile/Scrum, Code Reviews, Requirements Analysis, Documentation, Production Support, Stakeholder Communication

Projects

Chat2SQL Code | *Java, TypeScript, Node.js, SQL, PostgreSQL, Docker*

- Built an AI-powered SQL development workspace supporting 10+ databases and accelerating query authoring, schema exploration, and structured data management through a unified Java, TypeScript, and Docker-based interface.

Big-Brain Live | *LLaMA-2, FAISS, PostgreSQL, GCP, REST APIs*

- Engineered a reliable cloud-based data ingestion and indexing pipeline processing 50K+ records, implementing incremental updates, metadata persistence, and performance tuning to reduce retrieval latency by 40% overall.

Linguistix | *Python, AWS, Docker, CI/CD*

- Developed and deployed a containerized RESTful data-processing API on AWS using CodePipeline and ECS Fargate with automated releases, validation checks, 87% accuracy, and dependable production-grade operational reliability.